Multivariate Calculus Lecture Notes
Section 17.2 – Stokes's Theorem

I. Idea

a. Recall the circulation-curl form of Green's Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D \operatorname{curl}_z \mathbf{F}\, dA,$$

where C is the counterclockwise-oriented curve defined by the boundary of the region D in the plane.

i. Let's look at the double integral on the right side of Green's Theorem from a 3D perspective.

ii. Since D lies in the 2D Euclidean plane, the outward-pointing unit normal vector to D is $\mathbf{k}$.

iii. Thus we have

$$\iint_D \operatorname{curl}_z \mathbf{F}\, dA = \iint_D (\operatorname{curl}(\mathbf{F}) \cdot \mathbf{k})\, dA = \iint_D \operatorname{curl}(\mathbf{F})\, d\mathbf{S}.$$

iv. Thus the double integral in Green's Theorem (and double integrals in general) can be viewed as surface integrals!

b. What if we have the surface S consisting of the top and four sides of a cube in 3D?

i. Graphically:

ii. Applying the left side of Green's Theorem to each of the five sides of the cube separately and adding up, we get

1. The circulation over most of the boundaries cancels out.
2. We're left with just the counterclockwise circulation around the bottom.
3. Denoting the bottom (which is the boundary of the surface S) via ∂S, we get $\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$.

iii. If we let S denote the full surface consisting of the four sides and top, then the five separate double integrals on the right side of Green's Theorem sum to the surface integral $\iint_S \operatorname{curl}(\mathbf{F})\, d\mathbf{S}$.

iv. Putting this all together, we get $\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S \operatorname{curl}(\mathbf{F})\, d\mathbf{S}$.

c. But there's nothing particularly special about having five sides of a cube as our surface; the same idea works for lots of surfaces in 3D, such as this one:

II. Statement of Stokes's Theorem

a. Stokes's Theorem: Let S be an oriented surface with boundary curve ∂S, oriented counterclockwise with respect to S, and let $\mathbf{F}$ be a vector field whose components

have continuous partial derivatives on an open region containing S. Then

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S \operatorname{curl}(\mathbf{F})\, d\mathbf{S}.$$

b. "Counterclockwise orientation" in 3D
   i. "Counterclockwise orientation" in 3D requires some qualifications, as what's counterclockwise depends on how you're looking at the surface.
   ii. What we mean by this is the following:
      1. The surface S must be oriented with outward-pointing normal vector $\mathbf{n}$.
      2. "Counterclockwise" means that if someone is walking along the boundary of the surface with their head pointed in the same direction as the normal vector, the surface is on their left.

c. Stokes's Theorem implies that $\iint_S \operatorname{curl}(\mathbf{F})\, d\mathbf{S}$ has the same value for all surfaces S that have ∂S as their boundary.
   i. So if you're trying to do such a surface integral, and S is too complicated, switch to an easier surface with the same boundary!
   ii. This should remind you of the path independence property for line integrals in a conservative vector field.

d. Like Green's Theorem, Stokes's Theorem says that the double integral over a surface of a derivative (the curl in this case) is equal to a single integral over the boundary of that surface.

III. Examples

a. Find $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where C is the curve defined by the intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$ and oriented counterclockwise when looking from above it, and $\mathbf{F} = -y^2\mathbf{i} + x\mathbf{j} + z^2\mathbf{k}$.
   i. Graphically:

   ii. We can define S to be the surface in the plane $y + z = 2$ enclosed by the curve C and then use Stokes's Theorem.
   iii. Parameterizing the surface
      1. The projection onto the xy-plane is the disk of radius 1, which is given by $x^2 + y^2 \leq 1$.
      2. The easiest way to parameterize this is with $x = r\cos\theta$, $y = r\sin\theta$.
      3. This gives the disk as $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$.
      4. Then $z = 2 - y = 2 - r\sin\theta$.
      5. So we have $G(r, \theta) = (r\cos\theta, r\sin\theta, 2 - r\sin\theta)$, for $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$.
   iv. Finding $\mathbf{N}$.
      1. We have $\mathbf{T}_r = \cos\theta\mathbf{i} + \sin\theta\mathbf{j} - \sin\theta\mathbf{k}$.
      2. Also, $\mathbf{T}_\theta = -r\sin\theta\mathbf{i} + r\cos\theta\mathbf{j} - r\cos\theta\mathbf{k}$.
      3. Thus

$$\mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos\theta & \sin\theta & -\sin\theta \\ -r\sin\theta & r\cos\theta & -r\cos\theta \end{vmatrix}$$

$= (-r\sin\theta\cos\theta + r\sin\theta\cos\theta)\mathbf{i} - (-r\cos^2\theta - r\sin^2\theta)\mathbf{j} + (r\cos^2\theta + r\sin^2\theta)\mathbf{k}$

$= r\mathbf{j} + r\mathbf{k}.$

4. We know this is outward-pointing because the positive $\mathbf{k}$ component means $\mathbf{N}$ always points up.

v. Next,

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^2 & x & z^2 \end{vmatrix} = (1+2y)\mathbf{k}.$$

vi. Then curl $\mathbf{F}(G(r,\theta)) = (1 + 2r\sin\theta)\mathbf{k}$.

vii. Then curl $\mathbf{F} \cdot \mathbf{N} = r + 2r^2\sin\theta$.

viii. Therefore, by Stokes's Theorem,

$$\oint_C \mathbf{F}\cdot\, d\mathbf{r} = \iint_S \text{curl}(\mathbf{F})\, d\mathbf{S} = \int_0^{2\pi}\int_0^1 (r + 2r^2\sin\theta)\, dr\, d\theta = \int_0^{2\pi}\left[\frac{r^2}{2} + \frac{2r^3}{3}\sin\theta\right]_0^1 d\theta$$

$$= \int_0^{2\pi}\left(\frac{1}{2} + \frac{2}{3}\sin\theta\right)d\theta = \left[\frac{\theta}{2} - \frac{2}{3}\cos\theta\right]_0^{2\pi} = \pi - \frac{2}{3} - \left(0 - \frac{2}{3}\right) = \pi.$$

b. Find the counterclockwise circulation of $\mathbf{F} = 2y\mathbf{i} + 3z\mathbf{j} + x\mathbf{k}$ around the base of the pyramid with vertices $(0,0,0)$, $(0,2,0)$, $(1,1,1)$, and $(1/2,1,5)$.

i. Graphically:

ii. Preliminary thoughts

1. Integrating around the base of the pyramid will require three line integrals.
2. Integrating over the surface of the pyramid will require four surface integrals.
3. However, Stokes's Theorem says that we can use any surface with the same boundary.
4. So let's choose the surface defined by the base of the pyramid.

iii. Parameterizing the surface

1. The surface is a part of a plane, so we'll need the equation of that plane.
   - For a plane equation, we need a vector normal to the plane.
   - Two vectors in the plane are $\mathbf{u} = \langle 0,2,0\rangle$ and $\langle 1,1,1\rangle$.
   - We have

$$\mathbf{u}\times\mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{vmatrix} = 2\mathbf{i} - 2\mathbf{k}.$$

   - Thus the plane equation is $2x - 2z = 0$, or $z = x$.
2. Projecting the triangle onto the xy-plane yields the following:

3. With $z = x$, a parameterization of the triangle will be $G(x, y, z) = (x, y, x)$, for $x \le y \le 2 - x$ and $0 \le x \le 1$.

iv. Find $\mathbf{N}$.
1. We have $\mathbf{T}_x = \langle 1, 0, 1 \rangle$ and $\mathbf{T}_y = \langle 0, 1, 0 \rangle$.
2. Thus $\mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} = -\mathbf{i} + \mathbf{k}$.
3. This vector does point up, as we can see by the positive $\mathbf{k}$ component.

v. $\text{curl}(\mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2y & 3z & x \end{vmatrix} = -3\mathbf{i} - \mathbf{j} - 2\mathbf{k}$

vi. $\text{curl}(\mathbf{F}) \cdot \mathbf{N} = 3 - 2 = 1$.

vii. Therefore, by Stokes's Theorem,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl}(\mathbf{F})\, d\mathbf{S} = \int_0^1 \int_x^{2-x} dA = \frac{1}{2}(2)(1) = 1,$$

where we evaluated the double integral simply by calculating the area of the region of integration.

Homework: Section 17.2 #1, 5, 6, 11, 12, 13, 34